



Objects rendered with (right) and without (left) shadows

## 5.3 Cameras

Cameras in 3ds Max are similar to real-world cameras. In addition to controlling what you see through viewports, they allow you to take virtual photographs, or renders, of

a scene. Cameras in 3ds Max have controls such as focal length and aperture as well as f-stop and shutter speed for creating effects such as depth of field and motion blur. 3ds Max cameras, however, don't use film, so shutter speed and f-stop do nothing to affect the exposure of the image.

### 5.3.1 Types of cameras

Although all cameras in 3ds Max share the same sets of attributes, you'll see two types of cameras listed when you choose Create > Cameras or select the Cameras tab from the Create panel. You will find as Free Camera and Target Camera; the only difference between these cameras is that the target camera is aimed at a target object. Free Camera is a basic camera that can be controlled by using translation and rotation. Target Camera always points to the target. This makes positioning the camera simply a matter of translating the camera and the aim point, rather than rotating the camera. This can be useful when the camera is following a moving object. If the aim point follows the object, the camera will too.

### 5.3.2 Focal length

The focal length of a lens determines the distance it takes for the lens to focus the image to a point.

Focal length and angle of view are set by using the camera's Parameters rollout. This is also where you set clipping planes.

The closer the focal point is to the lens, the wider the angle of view. For example, an extremely short focal length creates a fish-eye effect. A standard wide-angle lens reduces this effect and a 50mm lens is one of the most common and is good for most general applications. Longer lenses can be good for portraits, but at very long lengths, the lens might flatten the scene too much. Shorter focal lengths mean that objects must be closer to the camera to fill the field of view.

#### Quick fact

The focal length of a lens determines the duration of the scene that can be focused, also called the field of view (FOV). In 3ds Max, these two parameters are directly linked: as the Lens option increases, the FOV option decreases.